
Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Industrial Revolution to 2024

DATA SOURCES

Global Carbon Budget 2025 • NASA

SCOPE

1750 – 2024

KEY INSIGHT

Current CO₂ level is the highest in at least 3 million years of Earth history

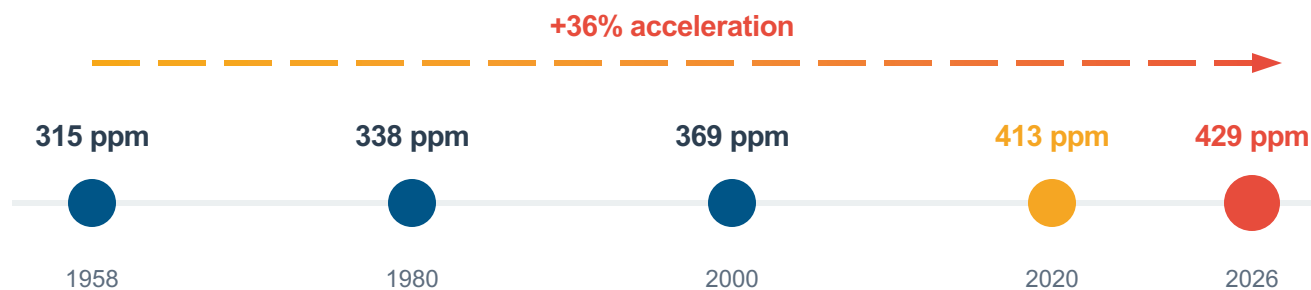
429

ppm

February 2026 — Mauna Loa Observatory

Pre-industrial baseline
~280 ppm

Acceleration Since 1958



KEY INSIGHT

Despite slowing growth rates, global emissions have not yet peaked

6

Billion Tonnes in 1950

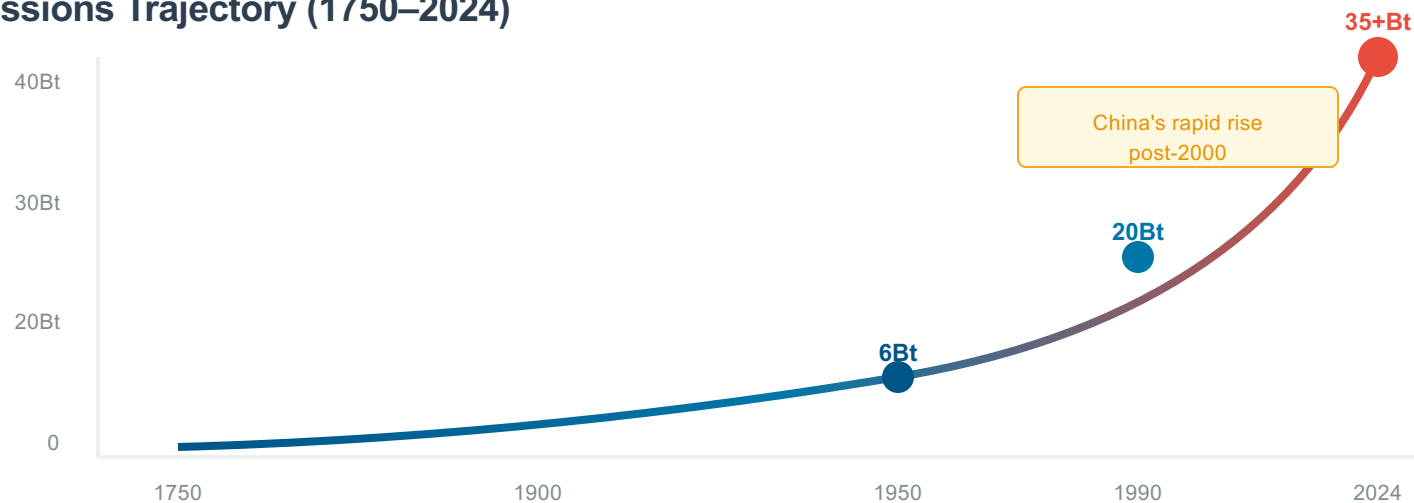
20

Billion Tonnes in 1990

35+

Billion Tonnes in 2024

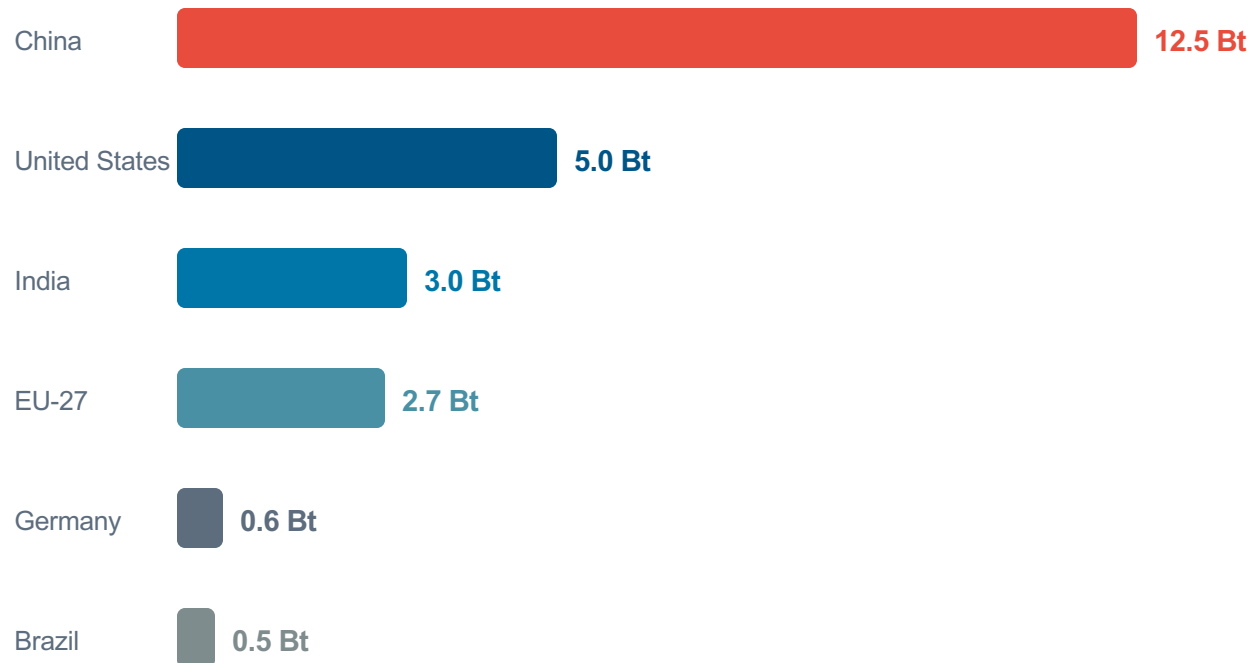
Emissions Trajectory (1750–2024)



KEY INSIGHT

China alone accounts for over 27% of global emissions — more than all EU-27 countries combined

Annual CO₂ Emissions by Country (Billion Tonnes)



China's Ascent

- 1950: ~0.1 Bt (negligible share)
- 1990: ~2.2 Bt (11% global)
- 2000: ~3.5 Bt (15% global)
- 2024: ~12.5 Bt (27% global)

US peak: ~6 Bt around 2005

Now declining to ~5 Bt

Global Share Context

1900: Europe + US > 90% of global emissions

1950: Europe + US still > 85%

2024: US + Europe < 33% combined

Asia now ~50% of global emissions

KEY INSIGHT

The geographic center of emissions has shifted dramatically in just over a century

1900

Europe + US dominated global emissions

>90%

Europe + United States

Rest of
World

Asia, Africa, South America — negligible contributions

2024

Asia leads; historic emitters decline

~50%

Asia

<33%

US + Europe

3-4%
Africa

3-4%
S. America

~5%
Other

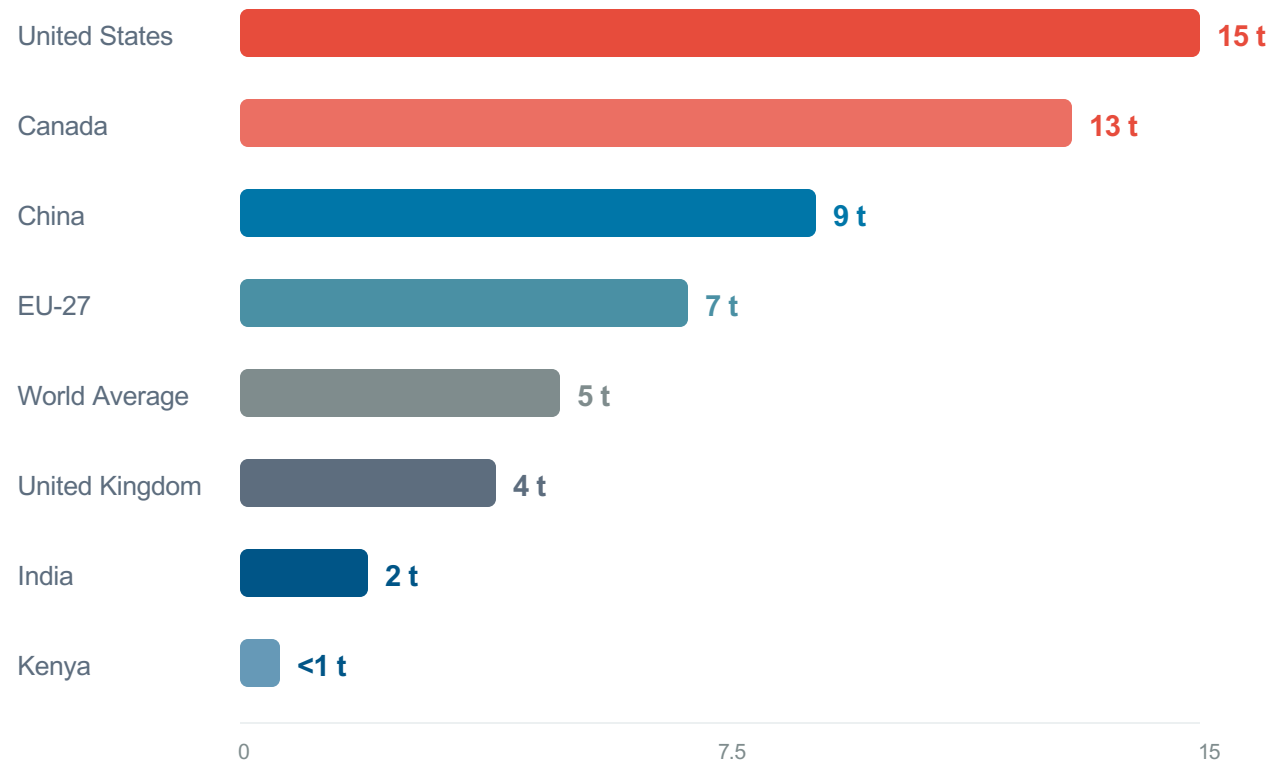
Africa and South America each contribute only 3–4% — comparable to international aviation and shipping combined

05 Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

KEY INSIGHT

The average American emits in under two days what the average person in Chad emits in a year

Per Capita CO₂ Emissions (tonnes/year)



150× DISPARITY

150×

Highest (US/Australia) vs.
Lowest (Chad, Niger)

Critical Context

- US per capita emissions are ~3× the global average
- High emitters: major oil nations (Qatar, UAE)
- Among large populations: US, Australia, Canada lead
- Chad/Niger: ~0.1 t/person (150× lower than US)

Policy implication: Per capita metrics reveal equity dimensions invisible in

KEY INSIGHT

Mixed signals: Declines in developed nations offset by rises in emerging economies

Regional Change Patterns

**DECLINING (-1% to -5%)**

- United States — sustained reduction
- Much of Europe — ongoing decarbonization

**RISING (+2% to +10%+)**

- Russia — significant increase
- Southeast Asia — rapid growth
- Parts of Africa — upward trajectory
- Some South American nations — very large % increases

2024 Annual Change Summary

- US emissions declined ~1–3%
- EU collectively reduced emissions
- China's emissions continued slight rise
- India emissions rising rapidly

Net global effect: Emissions still rising

Why This Matters

- US/Europe declines are real but insufficient
- Emerging economies prioritizing development
- Global 2024 emissions not yet declining
- Peak global emissions remain elusive

The transition is underway but incomplete

KEY INSIGHT

Countries with similar living standards can have very different per capita emissions based on energy sources

France & UK — Lower Per Capita

France

- ~70% nuclear electricity
- ~17% renewables
- ~5% fossil fuels in electricity

Per Capita: ~4 t CO₂

United Kingdom

- ~40% renewables + nuclear
- Significant coal phase-out
- Natural gas supplement

Per Capita: ~4 t CO₂

Combined: ~4 t per capita average

2x

Germany & Netherlands — Higher Per Capita

Germany

- ~50% fossil fuels (coal + gas)
- ~45% renewables
- Nuclear phase-out completed 2023

Per Capita: ~8 t CO₂

Netherlands

- Heavy reliance on natural gas
- Limited renewables penetration
- High industrial demand

Per Capita: ~9 t CO₂

Combined: ~8.5 t per capita average

KEY TAKEAWAYS

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1

35+ Billion Tonnes & Growing

Global emissions exceed 35 Bt and have not yet peaked

2

China: >25% of Global Total

~12.5 Bt — more than double the US (~5 Bt); India rising fast (~3 Bt)

3

150× Per Capita Disparity

US (~15 t) vs Chad/Niger (~0.1 t)
— equity dimensions invisible in totals

4

Shifting Geographic Center

Asia ~50% today; US+Europe <33%
vs. >90% combined in 1900

5

Energy Mix Determines Per Capita Outcomes

France/UK (~4 t) vs Germany/Netherlands (~8-9 t) — same prosperity, different choices.
Policy and infrastructure decisions shape emissions trajectories.

For Policy Analysts & Climate Negotiators:

Effective climate policy must address both total emissions AND per capita equity — recognizing that historical responsibility, current capacity, and energy choices all shape the path forward.